Soham Panchal

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Education

University of Texas at Arlington **Expected May 2028**
Honors B.S. in Computer Science · Minor in Mathematics · GPA: 3.75 *Arlington, TX*

Experience

Software Developer Intern **Jun 2026 – Aug 2026**
E-Sutra Technologies *Mumbai, India*

- Took ownership of stabilizing production backend workflows, tracing defects across REST APIs, authentication, SQL queries, relational data, and request/response handling to resolve failures at their source.
- Strengthened backend reliability through request validation, centralized error handling, and data-integrity checks, resolving server exceptions, inconsistent API responses, missing records, and invalid relational states.
- Performed end-to-end root-cause analysis and regression testing across application and database layers, collaborating with developers and QA to validate fixes before production release.

Projects

CampusAI – AI Academic Productivity Platform | *Next.js, FastAPI, PostgreSQL, pgvector, Supabase* **May 2026 – Sep 2026**

- Built and deployed a full-stack RAG platform that converts uploaded course PDFs into searchable study material using page-aware chunking, semantic embeddings, PostgreSQL/pgvector, and document-grounded question answering with source context.
- Implemented vector search, hybrid semantic/keyword retrieval with reciprocal rank fusion, and Cross-Encoder reranking; evaluated retrieval on 42 labeled questions and achieved 85.71% Recall@1, 97.62% Recall@5, and 0.900 MRR.
- Developed asynchronous PDF ingestion, course-scoped document management, flashcard and quiz generation, mastery tracking, and authenticated user ownership controls across a Next.js frontend and Python/FastAPI AI service.
- Built automated API, retrieval, embedding, PDF-processing, authentication, and generation tests with GitHub Actions CI, and designed a lower-memory production retrieval path for deployment under constrained cloud resources.

Hotel Management Platform | *React Native, Express, PostgreSQL, Prisma, Docker* **Jan 2026 – May 2026**

- Engineered a full-stack hotel operations platform with React Native and Express, implementing JWT authentication, Manager/Front Desk RBAC, reservations, guest management, billing, reporting, and role-aware mobile workflows.
- Implemented date-aware availability and interval-overlap validation with PostgreSQL and Prisma, preventing conflicting and blocked-room reservations while executing reservation creation inside database transactions for atomicity.
- Migrated persistence from SQLite to PostgreSQL and introduced Prisma migrations, repository/service architecture, Zod validation, centralized error handling, repeatable seeding, and PostgreSQL-backed Vitest/Supertest integration tests.
- Integrated Stripe Checkout with signature-verified webhooks for payment-state synchronization and deployed the Dockerized API on Render with Supabase PostgreSQL and GitHub Actions CI.

RideWise – ML Ride Decision Platform | *Java, FastAPI, XGBoost, Cloud Run* **Oct 2025 – Dec 2025**

- Built and deployed an Android ride-decision platform combining Google Routes traffic data, NYC geospatial processing, Firebase, and FastAPI services to generate historical ML fare estimates and route-aware ride recommendations.
- Trained separate XGBoost regression pipelines on approximately 1.5M NYC TLC high-volume for-hire trip records, achieving validation R^2 of 0.835 for Uber and 0.894 for Lyft with MAEs of $4.90 and $3.15.
- Designed Wait & Save and Walk Nearby algorithms that evaluate future departure windows and alternate pickup locations using traffic-aware driving/walking routes, taxi-zone resolution, and provider-specific fare models, with graceful fallback outside NYC.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C/C++
Frameworks & Libraries: Next.js, React, React Native, Node.js, Express.js, FastAPI, Spring Boot, XGBoost, scikit-learn, Prisma
Data & AI: PostgreSQL, pgvector, MySQL, Supabase, Firebase, OpenAI API
Testing & DevOps: Pytest, Vitest, Supertest, GitHub Actions, Docker, Postman
Cloud & APIs: Google Cloud Run, Render, Stripe, Google Maps Platform
Concepts: Data Structures & Algorithms, Machine Learning, RAG, Vector Search, API Design, Database Transactions, Relational Database Design